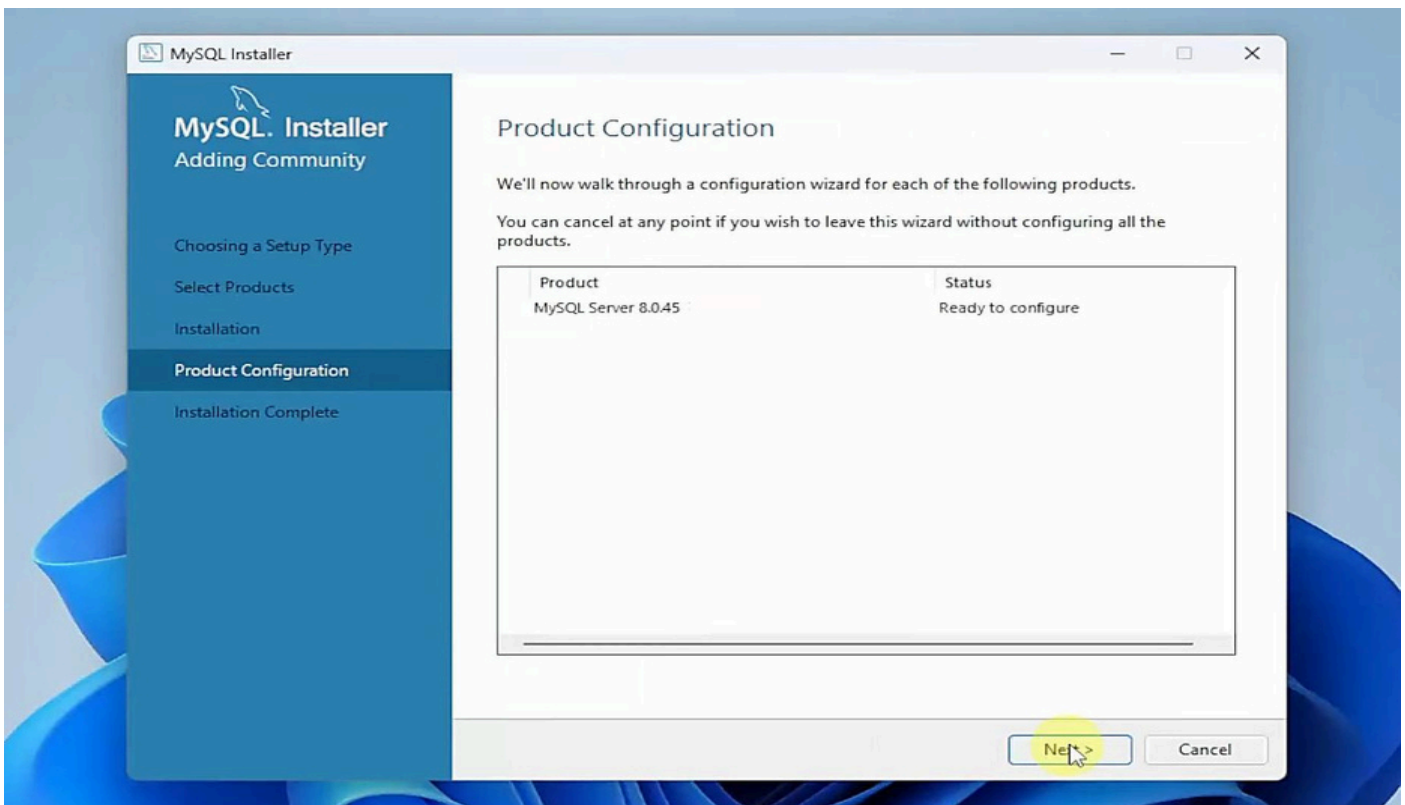
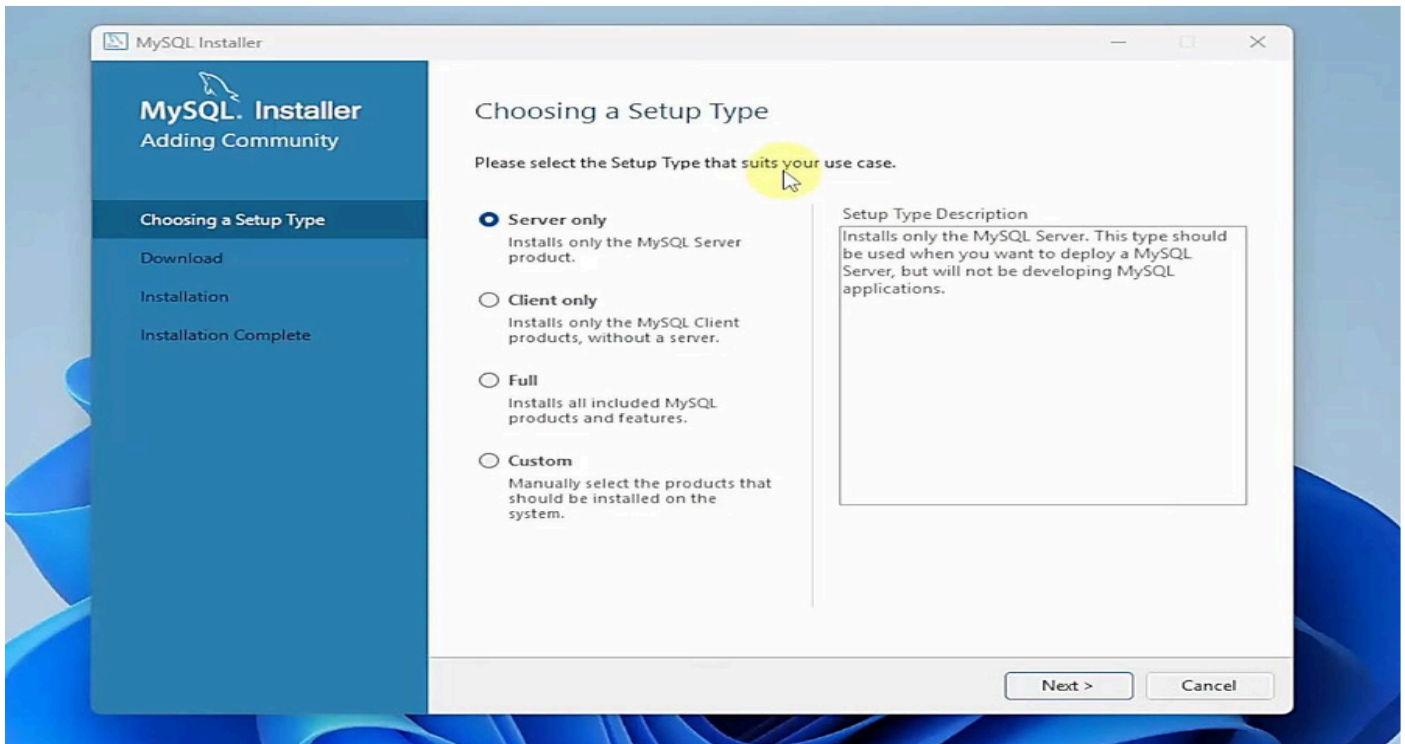
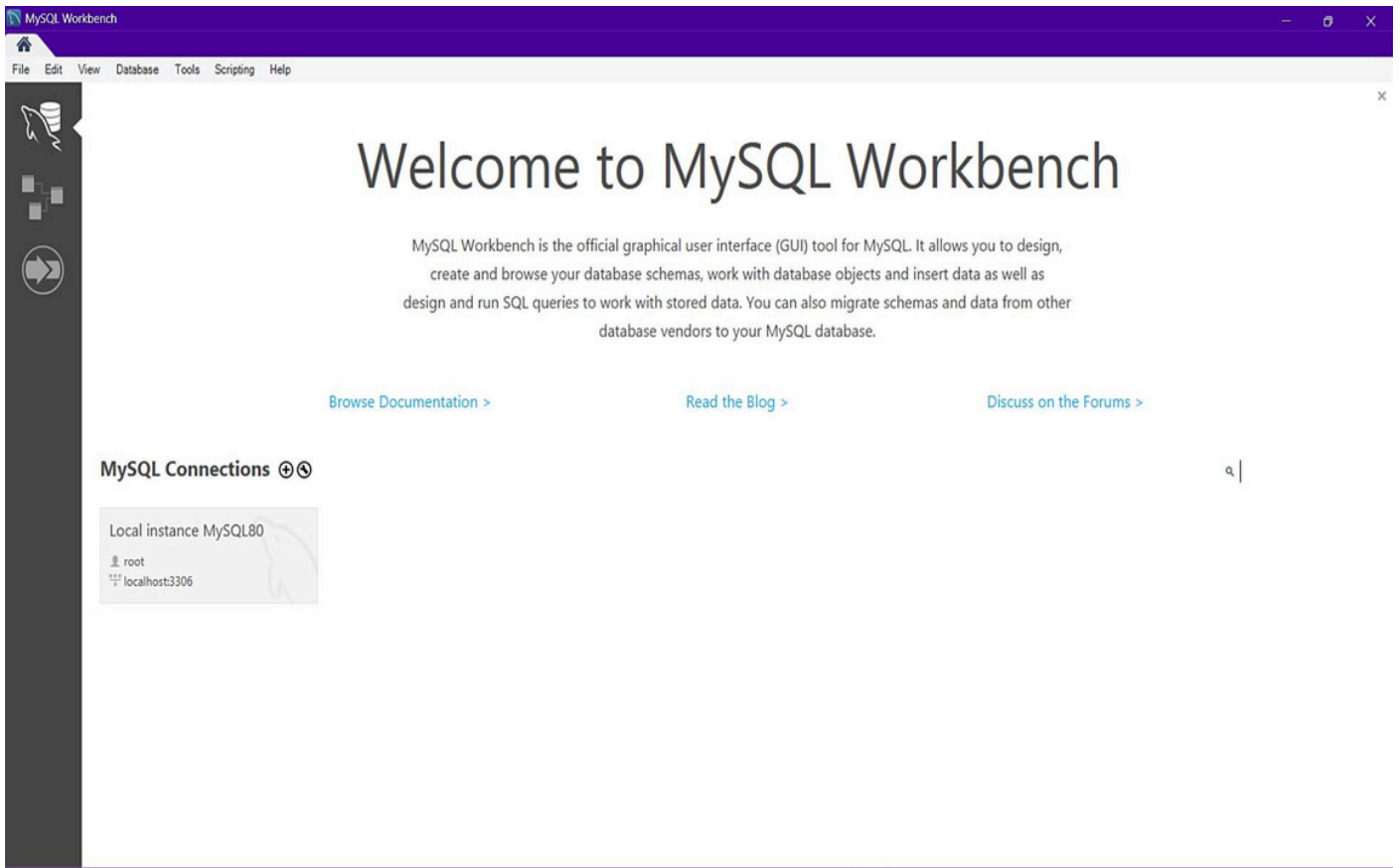
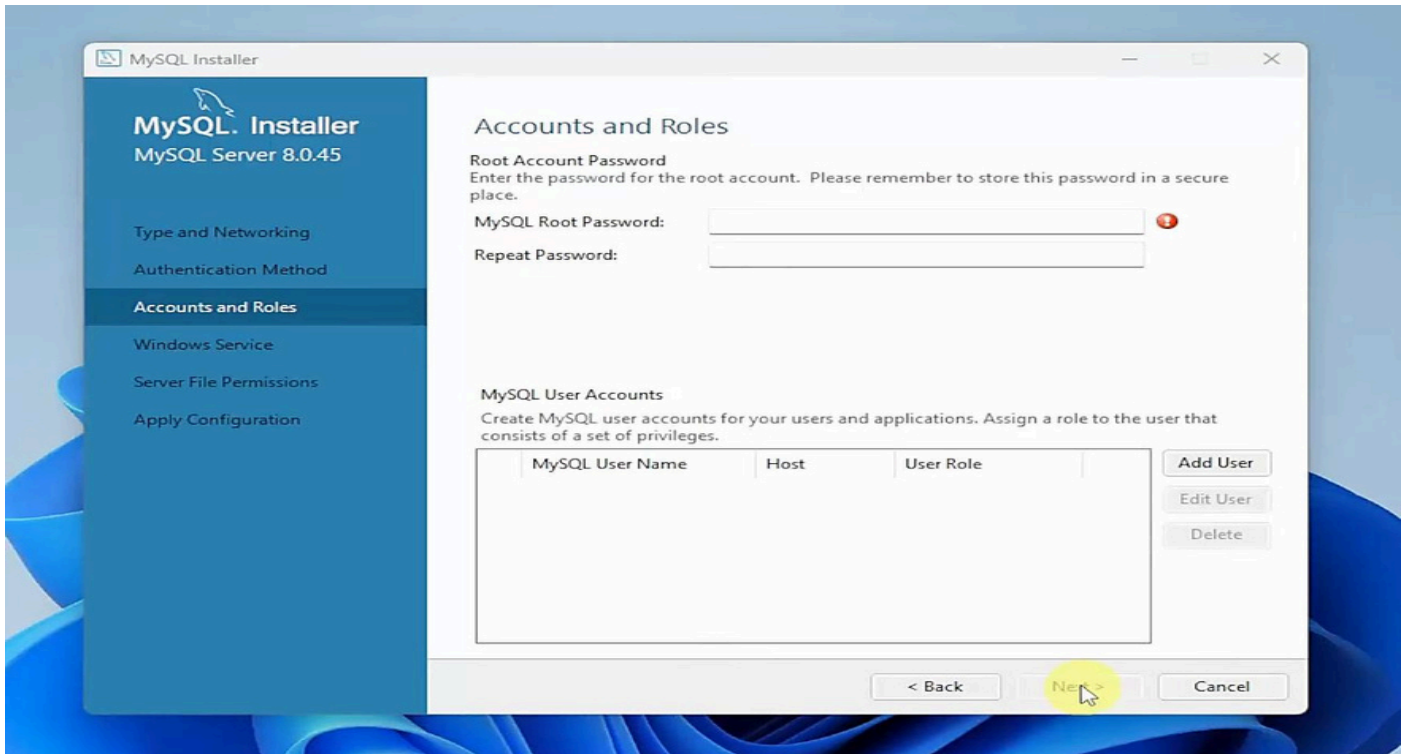


Practical 1





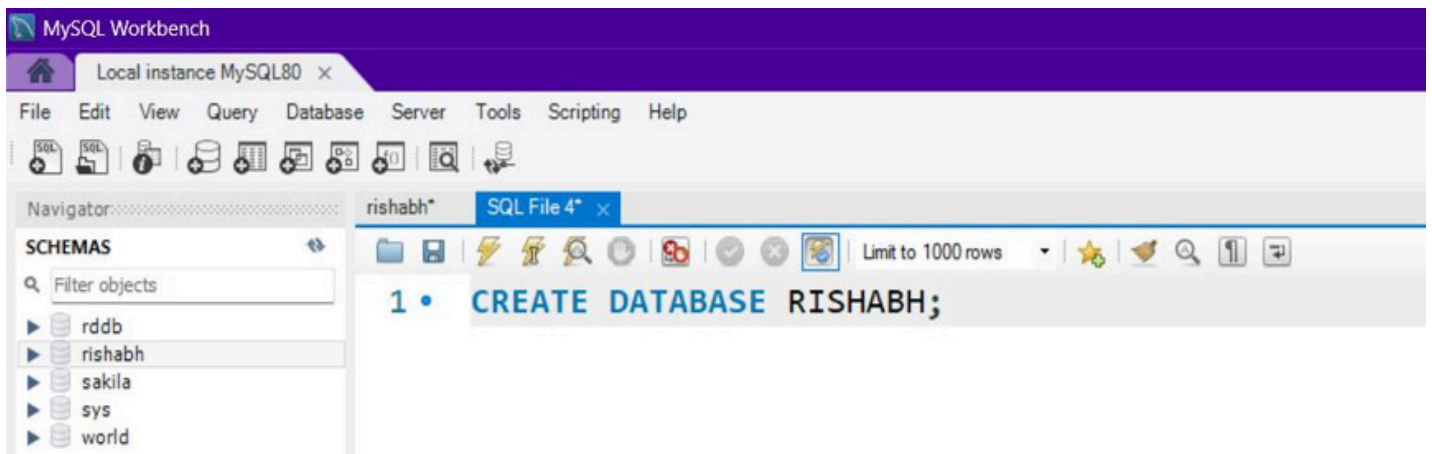
Practical 2

Aim: Perform the following:

- Creating a Database
- Viewing all databases,
- Creating Tables (With and Without Constraints)
- Viewing all Tables in a Database,
- Inserting/Updating/Deleting Records in a Table,
- Save point, Saving (Commit) and Undoing (rollback).

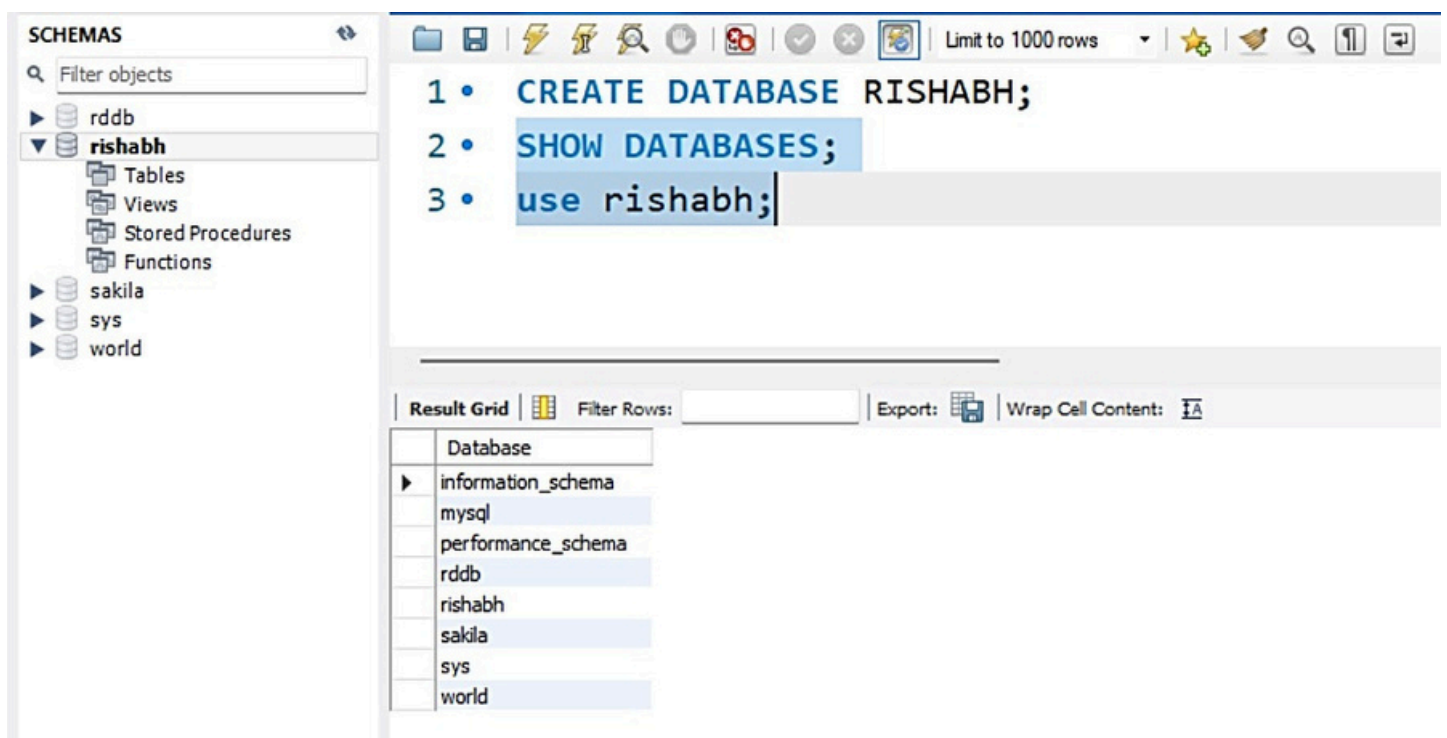
Source-Code & Output:

CREATE DATABASE RISHABH;



SHOW DATABASES;

use RISHABH;



```
CREATE TABLE Student (
    SID INT,
    Sname VARCHAR(20),
    Branch VARCHAR(20),
    city VARCHAR(20)
);
DESC Student;
```

Result Grid


Filter Rows:

Export:


	Field	Type	Null	Key	Default	Extra
▶	SID	int	YES		NULL	
	Sname	varchar(20)	YES		NULL	
	Branch	varchar(20)	YES		NULL	
	city	varchar(20)	YES		NULL	

```
INSERT INTO Student VALUE(1,'Rishabh Dwivedi','MCA','Nagpur');
INSERT INTO Student VALUE(2,'Aman Singh','MCA','Nagpur');
INSERT INTO Student VALUE(3,'Vijay Thakur','MCA','Nagpur');
SELECT * FROM Student;
```

	SID	Sname	Branch	city
►	1	Rishabh Dwivedi	MCA	Nagpur
	2	Aman Singh	MCA	Nagpur
	3	Vijay Thakur	MCA	Nagpur

```
UPDATE Student
SET city = 'Pune'
WHERE SID = 2;
SELECT * FROM Student;
```

	SID	Sname	Branch	city
►	1	Rishabh Dwivedi	MCA	Nagpur
	2	Aman Singh	MCA	Pune
	3	Vijay Thakur	MCA	Nagpur

```
DELETE FROM Student
WHERE SID = 3;
SELECT * FROM Student;
```

	SID	Sname	Branch	city
▶	1	Rishabh Dwivedi	MCA	Nagpur
	2	Aman Singh	MCA	Pune

```
CREATE TABLE Student_Constrained (
    SID INT PRIMARY KEY,
    Sname VARCHAR(20) NOT NULL,
    Branch VARCHAR(20),
    city VARCHAR(20)
);
DESC Student_Constrained;
```

	Field	Type	Null	Key	Default	Extra
▶	SID	int	NO	PRI	NULL	
	Sname	varchar(20)	NO		NULL	
	Branch	varchar(20)	YES		NULL	
	city	varchar(20)	YES		NULL	

```
START TRANSACTION;
SAVEPOINT sp1;
INSERT INTO Student VALUES (5, 'Shubham Dwivedi', 'BCCA', 'Nagpur');
ROLLBACK TO sp1;
SELECT * FROM Student;
Commit
```

	SID	Sname	Branch	city
▶	1	Rishabh Dwivedi	MCA	Nagpur
	2	Aman Singh	MCA	Pune

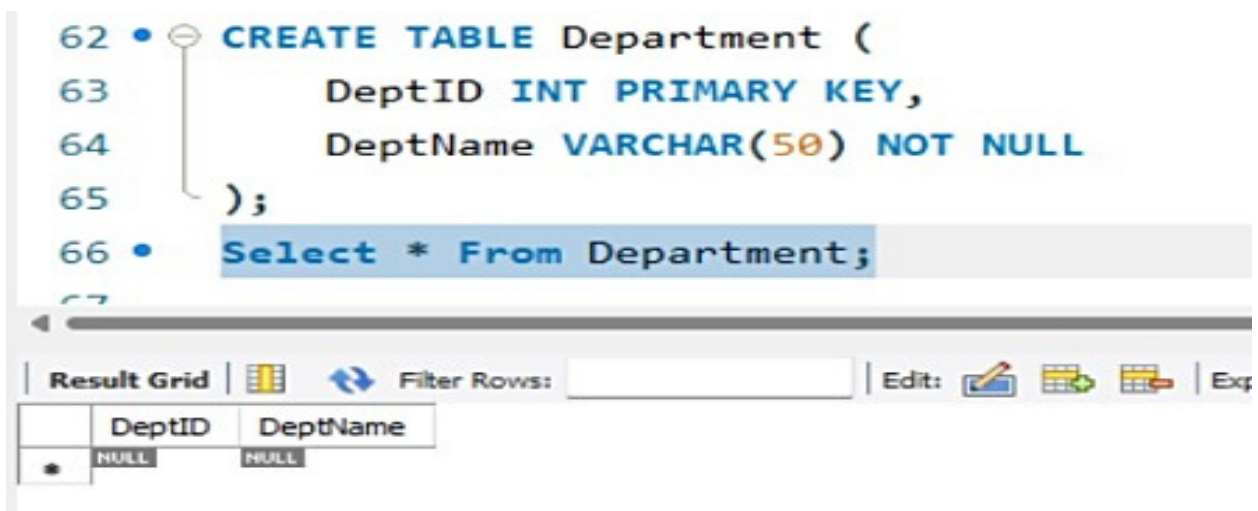
Practical 3

Perform the following:

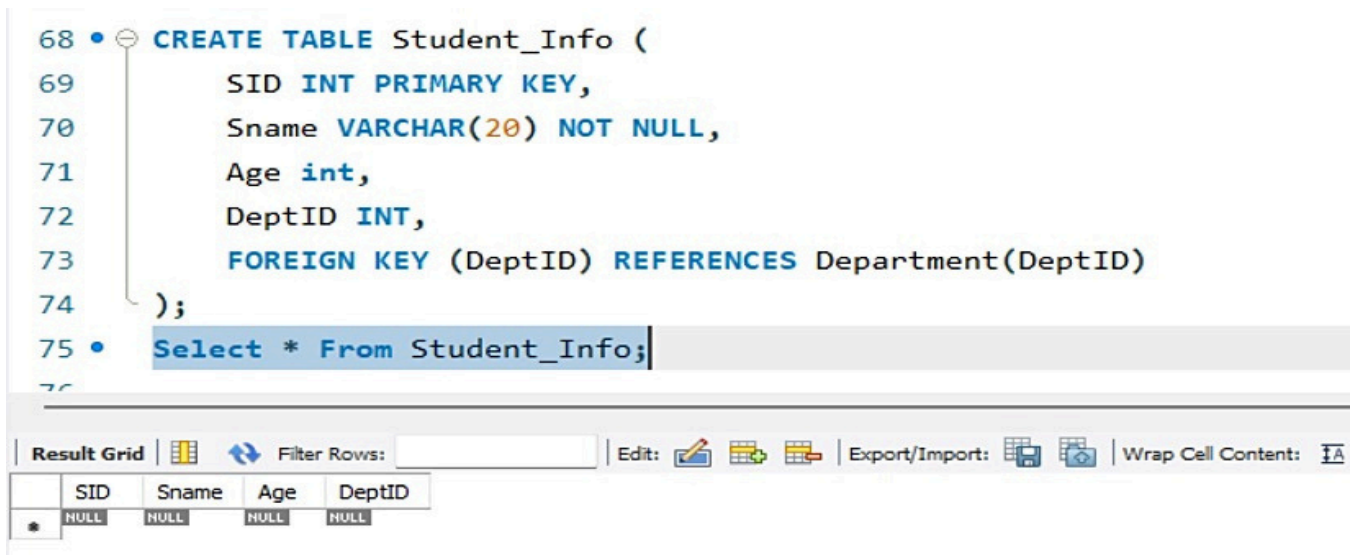
Data constraints (Primary key, Foreign key, Not Null) Data insertion into a table.

Source-Code & Output:

```
CREATE TABLE Department (  
    DeptID INT PRIMARY KEY,  
    DeptName VARCHAR(50) NOT NULL  
);  
Select * From Department;
```



```
CREATE TABLE Student_Info (  
    SID INT PRIMARY KEY,  
    Sname VARCHAR(20) NOT NULL,  
    Age INT,  
    DeptID INT,  
    FOREIGN KEY (DeptID) REFERENCES Department (DeptID)  
);  
SELECT * FROM Student_Info;
```




```
INSERT INTO Department VALUE(1,'MCA');
```

```
INSERT INTO Department VALUE(2,'MCA');
```

```
SELECT * FROM Department;
```

```
76 • INSERT INTO Department Value(1, 'MCA');
```

```
77 • INSERT INTO Department Value(2, 'MCA');
```

```
78 • Select * From Department;
```

Result Grid	Filter Rows:	Edit:	Export/Import
DeptID	DeptName		
1	MCA		
2	MCA		
NULL	NULL		

```
INSERT INTO Student_Info VALUE(10,'Pranit',22,1);
```

```
INSERT INTO Student_Info VALUE(20,'Rishabh',23,2);
```

```
SELECT * FROM Student_Info;
```

```
79 • INSERT INTO Student_Info Value(10, 'Pranit', '22', '1');
```

```
80 • INSERT INTO Student_Info Value(11, 'Rishabh', '23', '2');
```

```
81 • Select * From Student_Info;
```

Result Grid	Filter Rows:	Edit:	Export/Import:	Wrap
SID	Sname	Age	DeptID	
10	Pranit	22	1	
11	Rishabh	23	2	
NULL	NULL	NULL	NULL	

Practical 4

Aim: Perform the following: a. Altering a Table Dropping/Truncating/Renaming Tables, Backup/Restoring a Database.

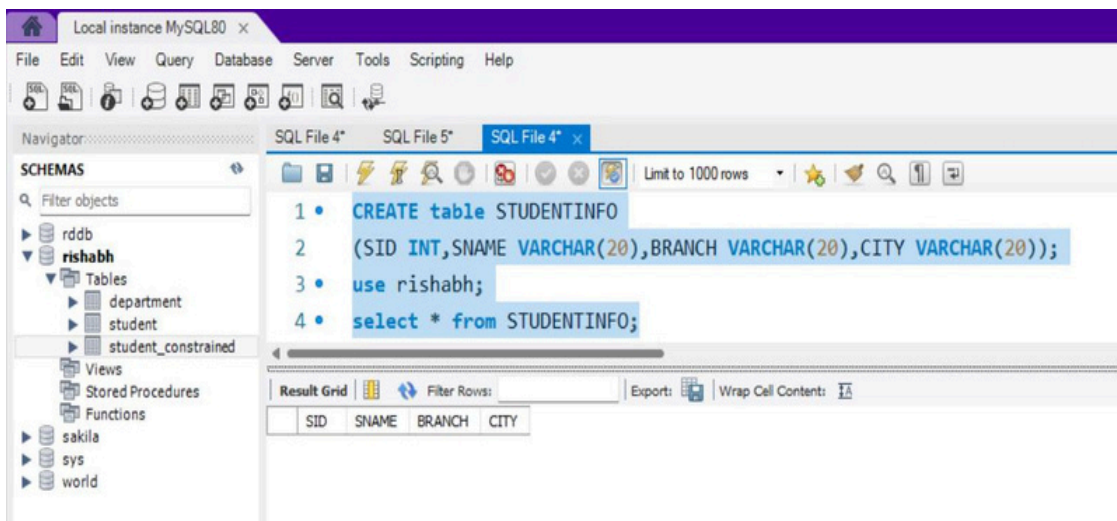
Source-code and Output:

```
CREATE table STUDENTINFO
```

```
(SID INT, SNAME VARCHAR(20), BRANCH VARCHAR(20), CITY VARCHAR(20));
```

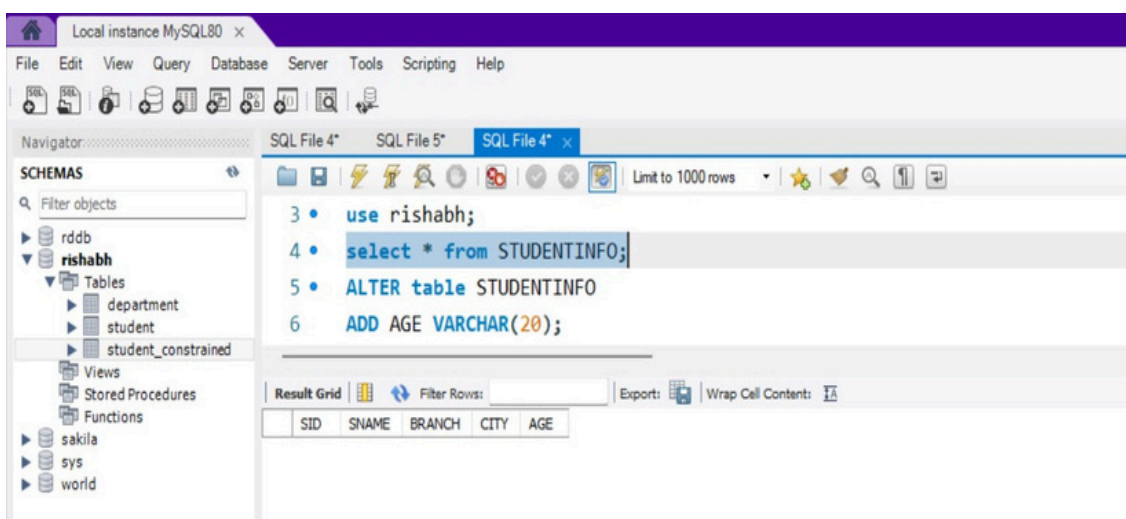
```
use rishabh;
```

```
select * from STUDENTINFO;
```



```
ALTER table STUDENTINFO
```

```
ADD AGE VARCHAR(20);
```

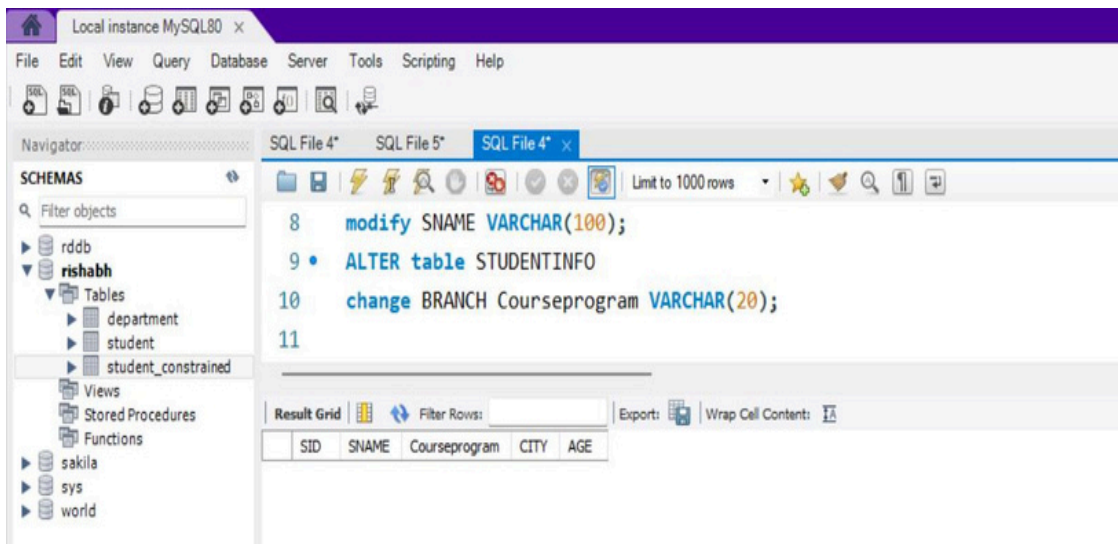


```
ALTER table STUDENTINFO
```

```
modify SNAME VARCHAR(100);
```

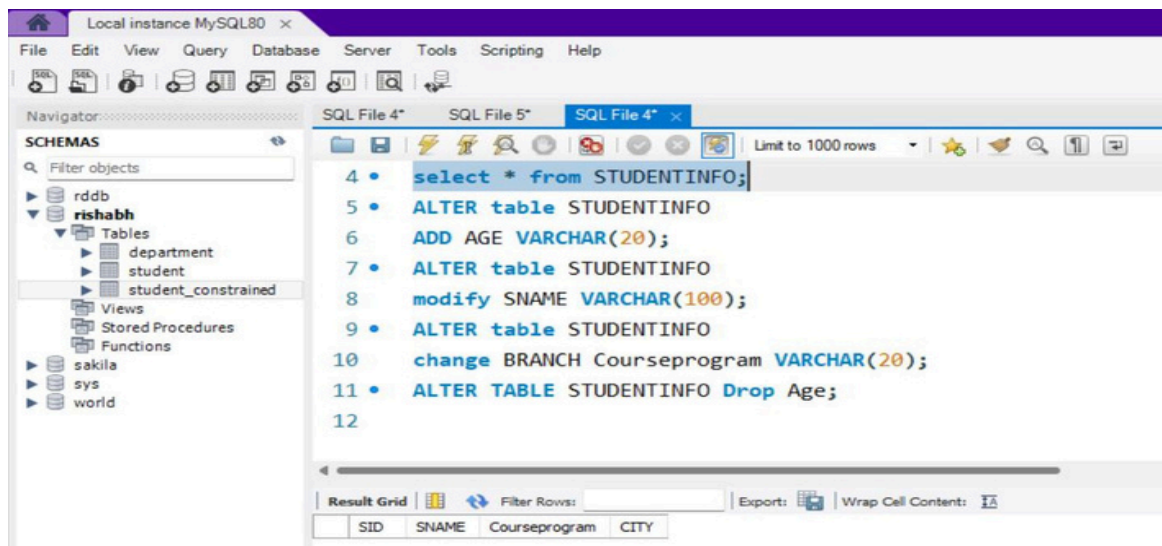

ALTER table STUDENTINFO

change BRANCH Courseprogram VARCHAR(20);



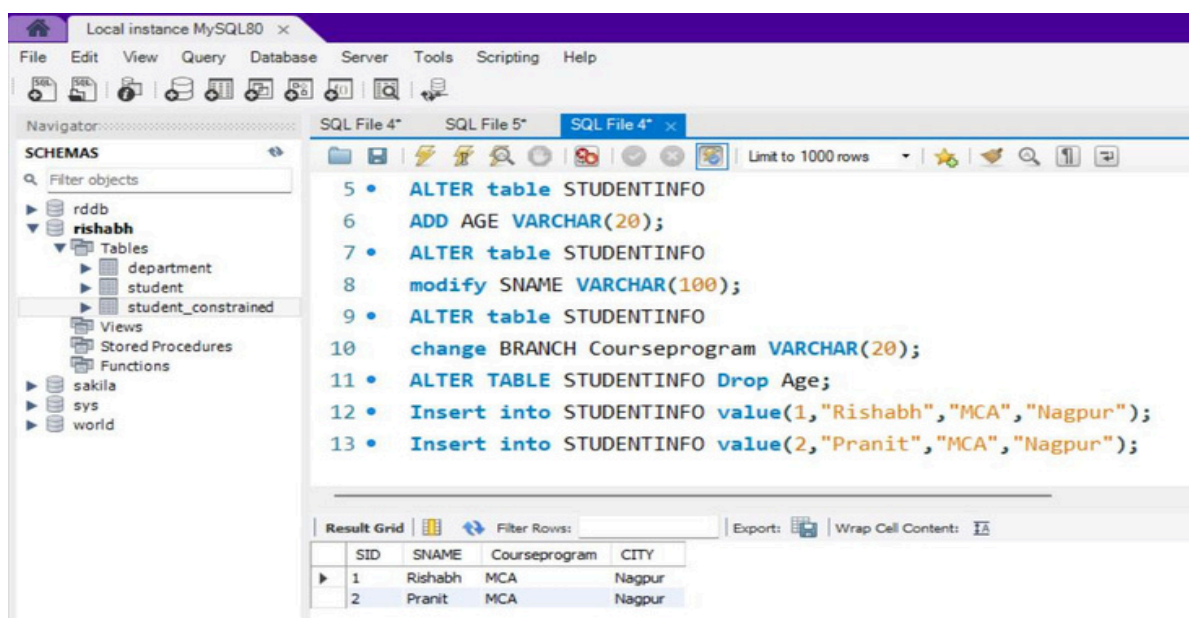
ALTER TABLE STUDENTINFO

Drop Age;



Insert into STUDENTINFO value(1,"Rishabh","MCA","Nagpur");

Insert into STUDENTINFO value(2,"Pranit","MCA","Nagpur");



rename table STUDENTINFO to Student_Info;

The screenshot shows the MySQL Workbench interface with the 'Schemas' tab selected. The 'Navigator' pane on the left shows the 'sakila' schema. The main editor displays a series of SQL queries. The 'Output' pane at the bottom shows the execution results of these queries.

```
1 • CREATE table STUDENTINFO
2 (SID INT,SNAME VARCHAR(20),BRANCH VARCHAR(20),CITY VARCHAR(20));
3 • use rishabh;
4 • select * from STUDENTINFO;
5 • ALTER table STUDENTINFO
6 ADD AGE VARCHAR(20);
7 • ALTER table STUDENTINFO
8 modify SNAME VARCHAR(100);
9 • ALTER table STUDENTINFO
10 change BRANCH Courseprogram VARCHAR(20);
11 • ALTER TABLE STUDENTINFO Drop Age;
12 • Insert into STUDENTINFO value(1,"Rishabh","MCA","Nagpur");
13 • Insert into STUDENTINFO value(2,"Pranit","MCA","Nagpur");
14 • rename table STUDENTINFO to Student_Info
```

#	Time	Action	Message
14	15:25:24	DROP TABLE Student_Info	Error Code: 1051. Unknown table 'rishabh.student_info'
15	15:28:58	Insert into STUDENTINFO value(1,"Rishabh","MCA","Nagpur")	1 row(s) affected
16	15:28:58	Insert into STUDENTINFO value(2,"Pranit","MCA","Nagpur")	1 row(s) affected
17	15:29:06	select * from STUDENTINFO LIMIT 0. 1000	2 row(s) returned
18	15:29:14	select * from STUDENTINFO LIMIT 0. 1000	2 row(s) returned
19	15:34:10	rename table STUDENTINFO to Student_Info	0 row(s) affected

truncate Student_info;

The screenshot shows the MySQL Workbench interface with the 'Schemas' tab selected. The 'Navigator' pane on the left shows the 'sakila' schema. The main editor displays a series of SQL queries. The 'Output' pane at the bottom shows the execution results of these queries.

```
8 • modify SNAME VARCHAR(100);
9 • ALTER table STUDENTINFO
10 change BRANCH Courseprogram VARCHAR(20);
11 • ALTER TABLE STUDENTINFO Drop Age;
12 • Insert into STUDENTINFO value(1,"Rishabh","MCA","Nagpur");
13 • Insert into STUDENTINFO value(2,"Pranit","MCA","Nagpur");
14 • rename table STUDENTINFO to Student_Info;
15 • truncate Student_info;
16 • select * from Student_info;
```

#	Time	Action	Message
15	15:28:58	truncate Student_info	0 row(s) affected
16	15:29:06	select * from Student_info	2 row(s) returned

drop table Student_info;

The screenshot shows the MySQL Workbench interface with the 'Schemas' tab selected. The 'Navigator' pane on the left shows the 'sakila' schema. The main editor displays a series of SQL queries. The 'Output' pane at the bottom shows the execution results of these queries.

```
3 • use rishabh;
4 • select * from STUDENTINFO;
5 • ALTER table STUDENTINFO
6 ADD AGE VARCHAR(20);
7 • ALTER table STUDENTINFO
8 modify SNAME VARCHAR(100);
9 • ALTER table STUDENTINFO
10 change BRANCH Courseprogram VARCHAR(20);
11 • ALTER TABLE STUDENTINFO Drop Age;
12 • Insert into STUDENTINFO value(1,"Rishabh","MCA","Nagpur");
13 • Insert into STUDENTINFO value(2,"Pranit","MCA","Nagpur");
14 • rename table STUDENTINFO to Student_Info;
15 • truncate Student_info;
16 • select * from Student_info;
17 • drop table Student_info;
```

#	Time	Action	Message
18	15:29:14	select * from STUDENTINFO LIMIT 0. 1000	2 row(s) returned
19	15:34:10	rename table STUDENTINFO to Student_Info	0 row(s) affected
20	15:37:24	truncate Student_info	0 row(s) affected
21	15:39:41	select * from STUDENTINFO LIMIT 0. 1000	Error Code: 1146. Table 'rishabh.studentinfo' doesn't exist
22	15:40:14	select * from Student_info LIMIT 0. 1000	0 row(s) returned
23	15:44:59	drop table Student_info	0 row(s) affected

Practical 5

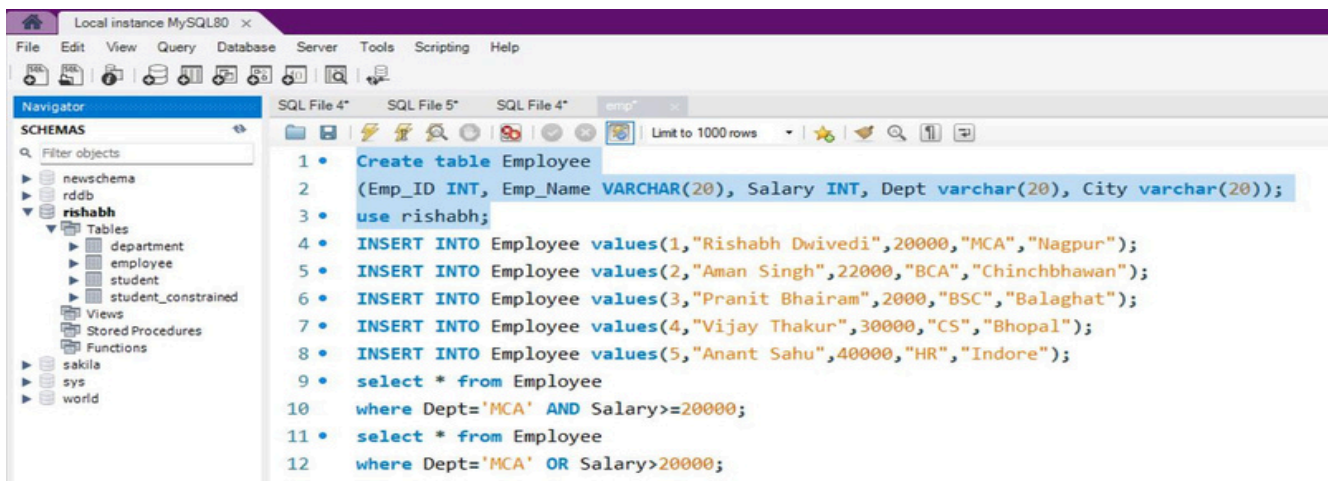
Aim: Implementation of different types of operators in SQL.

Source-Code & Output:

Create table Employee

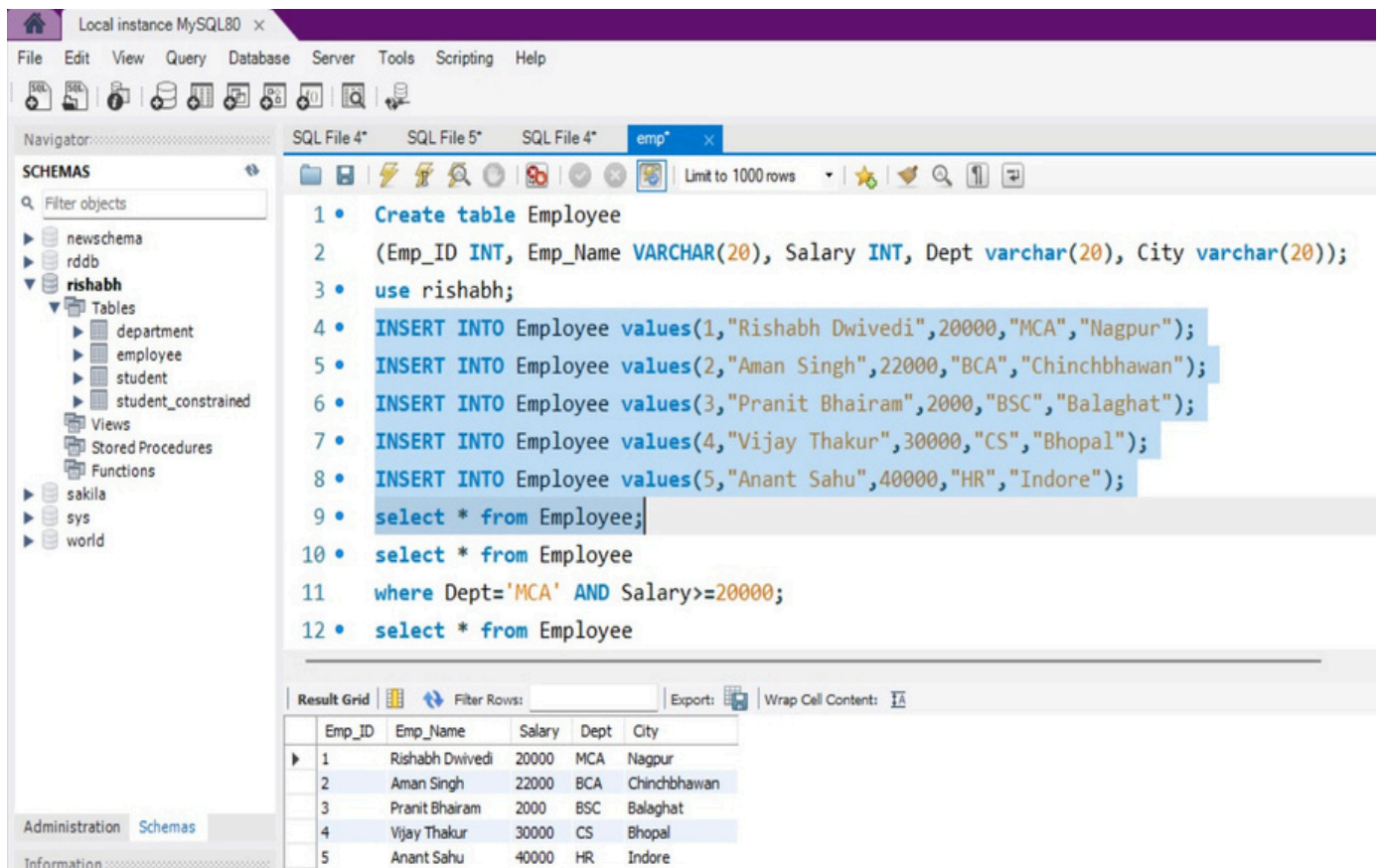
(Emp_ID INT, Emp_Name VARCHAR(20), Salary INT, Dept varchar(20), City varchar(20));

use rishabh;



```
1 • Create table Employee
2 (Emp_ID INT, Emp_Name VARCHAR(20), Salary INT, Dept varchar(20), City varchar(20));
3 • use rishabh;
4 • INSERT INTO Employee values(1,"Rishabh Dwivedi",20000,"MCA","Nagpur");
5 • INSERT INTO Employee values(2,"Aman Singh",22000,"BCA","Chinchbhawan");
6 • INSERT INTO Employee values(3,"Pranit Bhairam",2000,"BSC","Balaghat");
7 • INSERT INTO Employee values(4,"Vijay Thakur",30000,"CS","Bhopal");
8 • INSERT INTO Employee values(5,"Anant Sahu",40000,"HR","Indore");
9 • select * from Employee
10 where Dept='MCA' AND Salary>=20000;
11 • select * from Employee
12 where Dept='MCA' OR Salary>20000;
```

```
INSERT INTO Employee values(1,"Rishabh Dwivedi",20000,"MCA","Nagpur");
INSERT INTO Employee values(2,"Aman Singh",22000,"BCA","Chinchbhawan");
INSERT INTO Employee values(3,"Pranit Bhairam",2000,"BSC","Balaghat");
INSERT INTO Employee values(4,"Vijay Thakur",30000,"CS","Bhopal");
INSERT INTO Employee values(5,"Anant Sahu",40000,"HR","Indore");
select * from Employee;
```



```
1 • Create table Employee
2 (Emp_ID INT, Emp_Name VARCHAR(20), Salary INT, Dept varchar(20), City varchar(20));
3 • use rishabh;
4 • INSERT INTO Employee values(1,"Rishabh Dwivedi",20000,"MCA","Nagpur");
5 • INSERT INTO Employee values(2,"Aman Singh",22000,"BCA","Chinchbhawan");
6 • INSERT INTO Employee values(3,"Pranit Bhairam",2000,"BSC","Balaghat");
7 • INSERT INTO Employee values(4,"Vijay Thakur",30000,"CS","Bhopal");
8 • INSERT INTO Employee values(5,"Anant Sahu",40000,"HR","Indore");
9 • select * from Employee;
10 • select * from Employee
11 where Dept='MCA' AND Salary>=20000;
12 • select * from Employee
```

Emp_ID	Emp_Name	Salary	Dept	City
1	Rishabh Dwivedi	20000	MCA	Nagpur
2	Aman Singh	22000	BCA	Chinchbhawan
3	Pranit Bhairam	2000	BSC	Balaghat
4	Vijay Thakur	30000	CS	Bhopal
5	Anant Sahu	40000	HR	Indore

```
select * from Employee
where Dept='MCA' AND Salary >= 20000;
```

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'SCHEMAS' tree with 'rishabh' selected. The main editor window contains the following SQL script:

```
3 • use rishabh;
4 • INSERT INTO Employee values(1,"Rishabh Dwivedi",20000,"MCA","Nagpur");
5 • INSERT INTO Employee values(2,"Aman Singh",22000,"BCA","Chinchbhawan");
6 • INSERT INTO Employee values(3,"Pranit Bhairam",2000,"BSC","Balaghat");
7 • INSERT INTO Employee values(4,"Vijay Thakur",30000,"CS","Bhopal");
8 • INSERT INTO Employee values(5,"Anant Sahu",40000,"HR","Indore");
9 • select * from Employee;
10 • select * from Employee
11 • where Dept='MCA' AND Salary >= 20000;
12 • select * from Employee
13 • where Dept='MCA' OR Salary > 20000;
14 • select * from Employee
```

The 'Result Grid' at the bottom shows the output of the selected query (line 11):

Emp_ID	Emp_Name	Salary	Dept	City
1	Rishabh Dwivedi	20000	MCA	Nagpur

```
select * from Employee
where Dept='MCA' OR Salary > 20000;
```

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'SCHEMAS' tree with 'rishabh' selected. The main editor window contains the following SQL script:

```
3 • use rishabh;
4 • INSERT INTO Employee values(1,"Rishabh Dwivedi",20000,"MCA","Nagpur");
5 • INSERT INTO Employee values(2,"Aman Singh",22000,"BCA","Chinchbhawan");
6 • INSERT INTO Employee values(3,"Pranit Bhairam",2000,"BSC","Balaghat");
7 • INSERT INTO Employee values(4,"Vijay Thakur",30000,"CS","Bhopal");
8 • INSERT INTO Employee values(5,"Anant Sahu",40000,"HR","Indore");
9 • select * from Employee;
10 • select * from Employee
11 • where Dept='MCA' AND Salary >= 20000;
12 • select * from Employee
13 • where Dept='MCA' OR Salary > 20000;
14 • select * from Employee
```

The 'Result Grid' at the bottom shows the output of the selected query (line 13):

Emp_ID	Emp_Name	Salary	Dept	City
1	Rishabh Dwivedi	20000	MCA	Nagpur
2	Aman Singh	22000	BCA	Chinchbhawan
4	Vijay Thakur	30000	CS	Bhopal
5	Anant Sahu	40000	HR	Indore

```
select * from Employee
where NOT Dept='MCA';
```

```
14 • select * from Employee
15 where NOT Dept='MCA';
16 • select * from Employee
17 where Salary Between 25000 and 35000;
18 • select * from Employee
19 where Emp_Name like 'R%D%';
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	Emp_ID	Emp_Name	Salary	Dept	City
▶	2	Aman Singh	22000	BCA	Chinchbawan
	3	Pranit Bhairam	2000	BSC	Balaghat
	4	Vijay Thakur	30000	CS	Bhopal
	5	Anant Sahu	40000	HR	Indore

```
select * from Employee
where Salary Between 25000 and 35000;
```

```
16 • select * from Employee
17 where Salary Between 25000 and 35000;
18 • select * from Employee
19 where Emp_Name like 'R%D%';
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	Emp_ID	Emp_Name	Salary	Dept	City
▶	4	Vijay Thakur	30000	CS	Bhopal

```
select * from Employee
where Emp_Name like 'R%D%';
```

```
18 • select * from Employee
19 where Emp_Name like 'R%D%';
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	Emp_ID	Emp_Name	Salary	Dept	City
▶	1	Rishabh Dwivedi	20000	MCA	Nagpur

```
select * from Employee
where Salary<30000;
```



```
20 • select * from Employee
21 where Salary<30000;
```

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	Emp_ID	Emp_Name	Salary	Dept	City
▶	1	Rishabh Dwivedi	20000	MCA	Nagpur
	2	Aman Singh	22000	BCA	Chinchbhawan
	3	Pranit Bhairam	2000	BSC	Balaghat

select * from Employee
where Dept='BCA';

```
22 • select * from Employee
23 where Dept='BCA';
24 • select * from Employee
25 where Salary>30000;
```

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	Emp_ID	Emp_Name	Salary	Dept	City
▶	2	Aman Singh	22000	BCA	Chinchbhawan

select * from Employee
where Salary>30000;

```
24 • select * from Employee
25 where Salary>30000;
```

Result Grid



Filter Rows:

Export:



Wrap Cell Content:



	Emp_ID	Emp_Name	Salary	Dept	City
▶	5	Anant Sahu	40000	HR	Indore

select Emp_Name,Salary,Salary+5000 As Increased_Salary from Employee;

```
26 • select Emp_Name,Salary,Salary+5000 As Increased_Salary from Employee;
```

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	Emp_Name	Salary	Increased_Salary
▶	Rishabh Dwivedi	20000	25000
	Aman Singh	22000	27000
	Pranit Bhairam	2000	7000
	Vijay Thakur	30000	35000
	Anant Sahu	40000	45000

select Emp_Name,Salary,Salary-1000 As Decreased_Salary from Employee;

27 • `select Emp_Name,Salary,Salary-1000 As Decreased_Salary from Employee;`

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
Emp_Name	Salary	Decreased_Salary	
Rishabh Dwivedi	20000	19000	
Aman Singh	22000	21000	
Pranit Bhairam	2000	1000	
Vijay Thakur	30000	29000	
Anant Sahu	40000	39000	

select Emp_Name,Salary,Salary*2 As Doubled_Salary from Employee;

28 • `select Emp_Name,Salary,Salary*2 As Doubled_Salary from Employee;`

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
Emp_Name	Salary	Doubled_Salary	
Rishabh Dwivedi	20000	40000	
Aman Singh	22000	44000	
Pranit Bhairam	2000	4000	
Vijay Thakur	30000	60000	
Anant Sahu	40000	80000	

select Emp_Name,Salary,Salary/2 As Half_Salary from Employee;

29 • `select Emp_Name,Salary,Salary/2 As Half_Salary from Employee;`

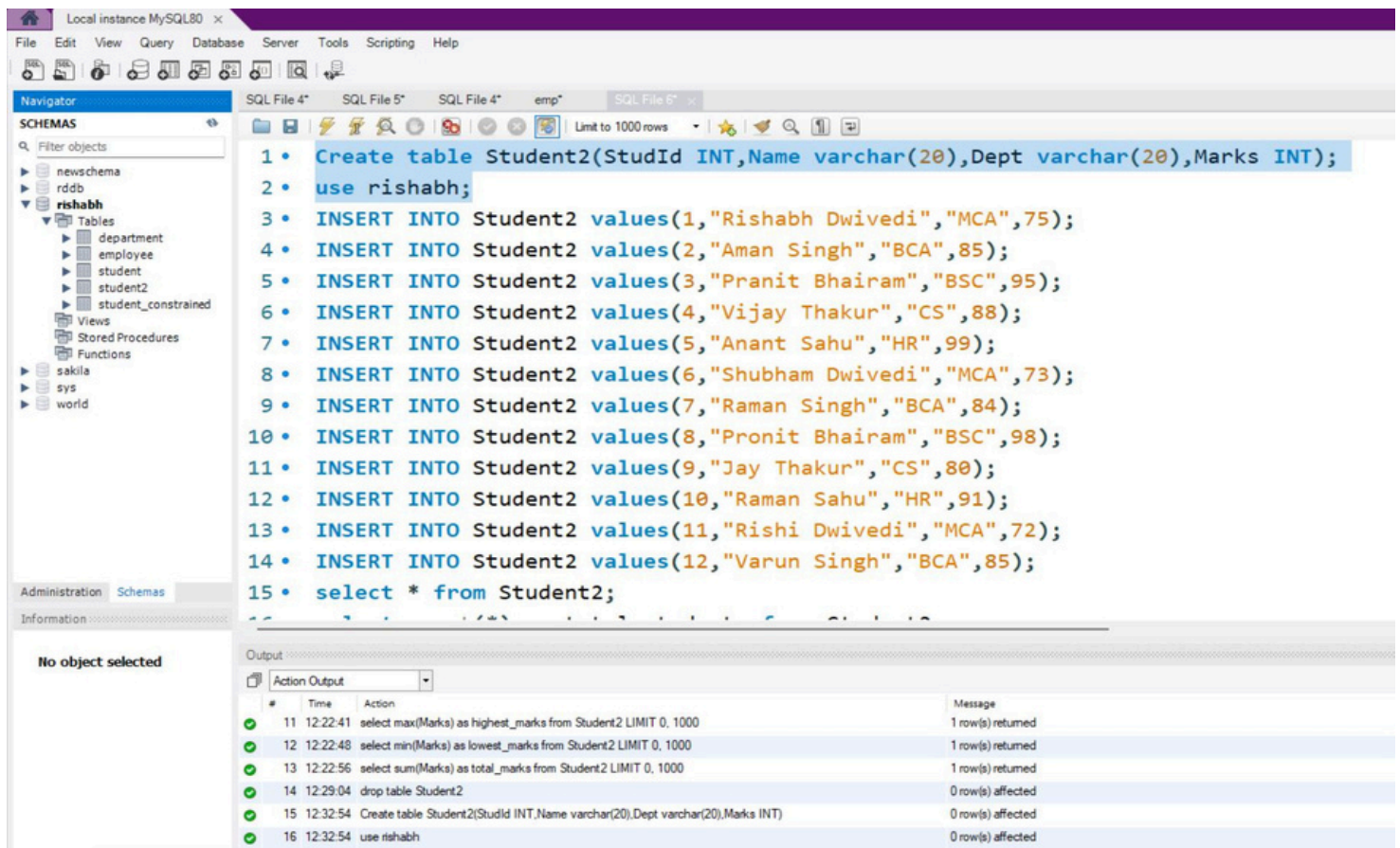
Result Grid	Filter Rows:	Export:	Wrap Cell Content:
Emp_Name	Salary	Half_Salary	
Rishabh Dwivedi	20000	10000.0000	
Aman Singh	22000	11000.0000	
Pranit Bhairam	2000	1000.0000	
Vijay Thakur	30000	15000.0000	
Anant Sahu	40000	20000.0000	

Practical 6

Aim: For a given set of relation schemes, create tables and perform the following Simple Queries, Simple Queries with Aggregate functions, Queries with Aggregate functions (group by and having clause).

Source-Code & Output:

Create table Student2(StudId INT,Name varchar(20),Dept varchar(20),Marks INT);
use rishabh;



The screenshot displays the MySQL Workbench interface. The left sidebar shows the 'SCHEMAS' tree with 'rishabh' selected. The main editor contains 15 SQL statements. The 'Output' pane at the bottom shows the execution results for statements 11 through 16.

```
1 • Create table Student2(StudId INT,Name varchar(20),Dept varchar(20),Marks INT);
2 • use rishabh;
3 • INSERT INTO Student2 values(1,"Rishabh Dwivedi","MCA",75);
4 • INSERT INTO Student2 values(2,"Aman Singh","BCA",85);
5 • INSERT INTO Student2 values(3,"Pranit Bhairam","BSC",95);
6 • INSERT INTO Student2 values(4,"Vijay Thakur","CS",88);
7 • INSERT INTO Student2 values(5,"Anant Sahu","HR",99);
8 • INSERT INTO Student2 values(6,"Shubham Dwivedi","MCA",73);
9 • INSERT INTO Student2 values(7,"Raman Singh","BCA",84);
10 • INSERT INTO Student2 values(8,"Pronit Bhairam","BSC",98);
11 • INSERT INTO Student2 values(9,"Jay Thakur","CS",80);
12 • INSERT INTO Student2 values(10,"Raman Sahu","HR",91);
13 • INSERT INTO Student2 values(11,"Rishi Dwivedi","MCA",72);
14 • INSERT INTO Student2 values(12,"Varun Singh","BCA",85);
15 • select * from Student2;
```

#	Time	Action	Message
11	12:22:41	select max(Marks) as highest_marks from Student2 LIMIT 0, 1000	1 row(s) returned
12	12:22:48	select min(Marks) as lowest_marks from Student2 LIMIT 0, 1000	1 row(s) returned
13	12:22:56	select sum(Marks) as total_marks from Student2 LIMIT 0, 1000	1 row(s) returned
14	12:29:04	drop table Student2	0 row(s) affected
15	12:32:54	Create table Student2(StudId INT,Name varchar(20),Dept varchar(20),Marks INT)	0 row(s) affected
16	12:32:54	use rishabh	0 row(s) affected

```
INSERT INTO Student2 values(1,"Rishabh Dwivedi","MCA",75);
INSERT INTO Student2 values(2,"Aman Singh","BCA",85);
INSERT INTO Student2 values(3,"Pranit Bhairam","BSC",95);
INSERT INTO Student2 values(4,"Vijay Thakur","CS",88);
INSERT INTO Student2 values(5,"Anant Sahu","HR",99);
INSERT INTO Student2 values(6,"Shubham Dwivedi","MCA",73);
INSERT INTO Student2 values(7,"Raman Singh","BCA",84);
INSERT INTO Student2 values(8,"Pronit Bhairam","BSC",98);
INSERT INTO Student2 values(9,"Jay Thakur","CS",80);
INSERT INTO Student2 values(10,"Raman Sahu","HR",91);
INSERT INTO Student2 values(11,"Rishi Dwivedi","MCA",72);
INSERT INTO Student2 values(12,"Varun Singh","BCA",85);
select * from Student2;
```


The screenshot shows the MySQL Workbench interface. The SQL editor contains the following queries:

```
1 • Create table Student2(StudId INT,Name varchar(20),Dept varchar(20),Marks INT);
2 • use rishabh;
3 • INSERT INTO Student2 values(1,"Rishabh Dwivedi","MCA",75);
4 • INSERT INTO Student2 values(2,"Aman Singh","BCA",85);
```

The result grid below shows the data inserted into the Student2 table:

StudId	Name	Dept	Marks
1	Rishabh Dwivedi	MCA	75
2	Aman Singh	BCA	85
3	Pranit Bhairam	BSC	95
4	Vijay Thakur	CS	88
5	Anant Sahu	HR	99
6	Shubham Dwivedi	MCA	73
7	Raman Singh	BCA	84
8	Pranit Bhairam	BSC	98
9	Jay Thakur	CS	80
10	Raman Sahu	HR	91
11	Rishi Dwivedi	MCA	72
12	Varun Singh	BCA	85

select count(*) as total_students from Student2;

The screenshot shows the MySQL Workbench interface with the following query and result:

```
16 • select count(*) as total_students from Student2;
```

The result grid shows:

total_students
12

select avg(Marks) as average_marks from Student2;

The screenshot shows the MySQL Workbench interface with the following query and result:

```
17 • select avg(Marks) as average_marks from Student2;
```

The result grid shows:

average_marks
85.4167

select max(Marks) as highest_marks from Student2;

The screenshot shows the MySQL Workbench interface with the following query and result:

```
18 • select max(Marks) as highest_marks from Student2;
```

The result grid shows:

highest_marks
99

select min(Marks) as lowest_marks from Student2;

The screenshot shows the MySQL Workbench interface with the following query and result:

```
19 • select min(Marks) as lowest_marks from Student2;
```

The result grid shows:

lowest_marks
72

select sum(Marks) as total_marks from Student2;

20 • `select sum(Marks) as total_marks from Student2;`

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	total_marks
▶	1025

select Dept, avg(Marks) as Avg_marks from Student2 group by Dept;

21 • `select Dept, avg(Marks) as Avg_marks from Student2 group by Dept;`

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	Dept	Avg_marks
▶	MCA	73.3333
	BCA	84.6667
	BSC	96.5000
	CS	84.0000
	HR	95.0000

select Dept, avg(Marks) as Avg_marks from Student2 group by Dept having avg(Marks)>80;

22 • `select Dept, avg(Marks) as Avg_marks from Student2 group by Dept having avg(Marks)>80;`

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	Dept	Avg_marks
▶	BCA	84.6667
	BSC	96.5000
	CS	84.0000
	HR	95.0000

Practical 7

Aim: Write a PL/SQL program using FOR loop to insert ten rows into a database table.

Source-Code & Output:

```
CREATE TABLE EMPLOYEE (  
    empId NUMBER PRIMARY KEY,  
    emp_name VARCHAR2(15) NOT NULL,  
    salary NUMBER  
);  
DECLARE  
    i NUMBER;  
BEGIN  
    FOR i IN 1..10  
    LOOP  
        INSERT INTO EMPLOYEE  
        VALUES(  
            i,  
            'EMPLOYEE_' || i, 20000 + (i * 1000)  
        );  
    END LOOP;  
    COMMIT;  
    DBMS_OUTPUT.PUT_LINE('10 rows inserted successfully');  
END;  
/
```

queries.sql + 44r3xwa6x

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```
1  -- Create table
2  CREATE TABLE EMPLOYEE (
3      empId NUMBER PRIMARY KEY,
4      emp_name VARCHAR2(15) NOT NULL,
5      salary NUMBER
6  );
7  DECLARE
8      i NUMBER;
9  BEGIN
10 FOR i IN 1..10
11 LOOP
12 INSERT INTO EMPLOYEE VALUES(i, 'EMPLOYEE_' || i, 20000 + (i * 1000));
13 END LOOP;
14 COMMIT;
15 DBMS_OUTPUT.PUT_LINE('10 rows inserted successfully');
16 END;
17 /
18 Select * from EMPLOYEE;
19
```

STDIN

Input for the program (Optional)

Output:

10 rows inserted successfully

	EMPID	EMP_NAME	SALARY
1	EMPLOYEE_1		21000
2	EMPLOYEE_2		22000
3	EMPLOYEE_3		23000
4	EMPLOYEE_4		24000
5	EMPLOYEE_5		25000
6	EMPLOYEE_6		26000
7	EMPLOYEE_7		27000
8	EMPLOYEE_8		28000
9	EMPLOYEE_9		29000
10	EMPLOYEE_10		30000

Practical 8

Name: Rishabh Dwivedi

Roll No: MC250068CA

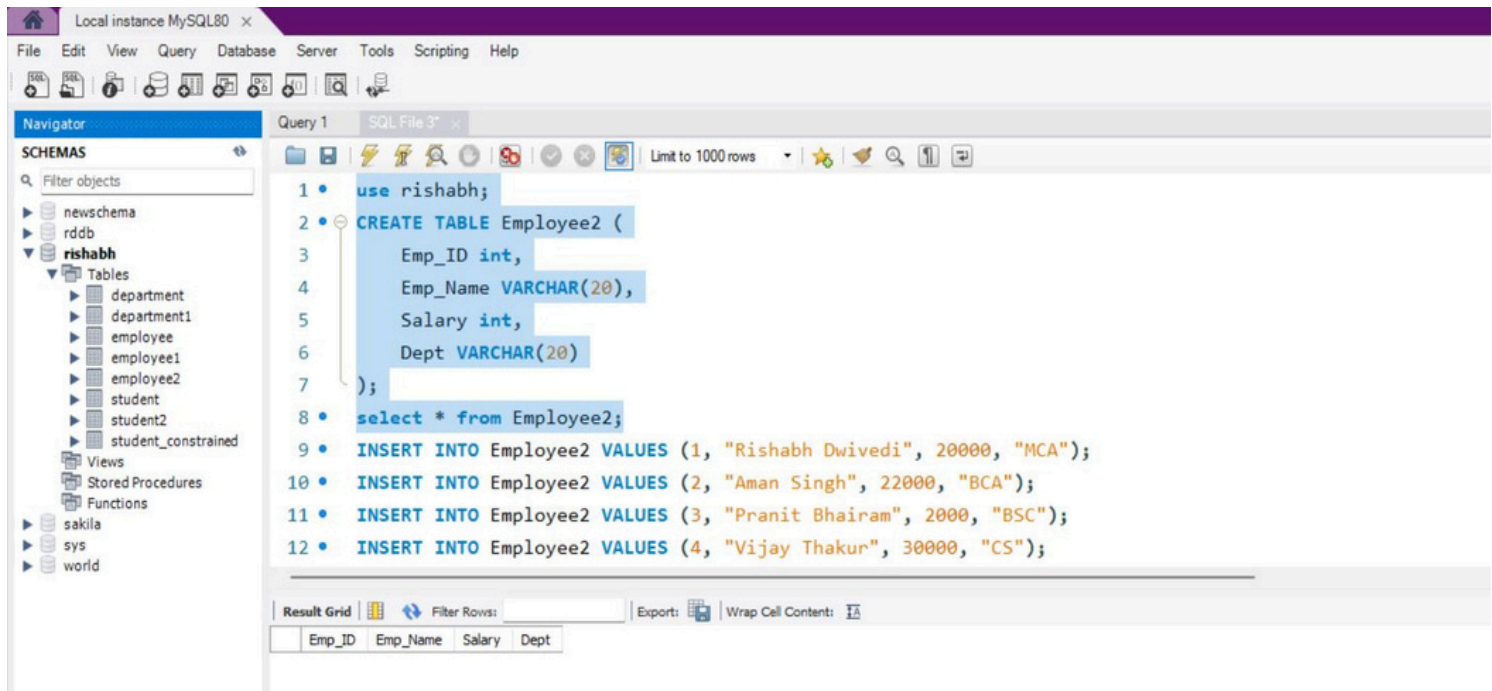
Aim: Writing SQL and PL/SQL queries to retrieve information from the databases.

Source-Code & Output:

create table Employee2;

use rishabh;

CREATE TABLE Employee2 (Emp_ID int, Emp_Name VARCHAR(20), Salary int, Dept VARCHAR(20));



The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'SCHEMAS' tree with the 'rishabh' database selected. The main editor window shows a SQL script for creating the 'Employee2' table and inserting four records. The table has columns: Emp_ID (int), Emp_Name (VARCHAR(20)), Salary (int), and Dept (VARCHAR(20)). The inserted records are: (1, 'Rishabh Dwivedi', 20000, 'MCA'), (2, 'Aman Singh', 22000, 'BCA'), (3, 'Pranit Bhairam', 2000, 'BSC'), and (4, 'Vijay Thakur', 30000, 'CS').

```
1 use rishabh;
2 CREATE TABLE Employee2 (
3     Emp_ID int,
4     Emp_Name VARCHAR(20),
5     Salary int,
6     Dept VARCHAR(20)
7 );
8 select * from Employee2;
9 INSERT INTO Employee2 VALUES (1, "Rishabh Dwivedi", 20000, "MCA");
10 INSERT INTO Employee2 VALUES (2, "Aman Singh", 22000, "BCA");
11 INSERT INTO Employee2 VALUES (3, "Pranit Bhairam", 2000, "BSC");
12 INSERT INTO Employee2 VALUES (4, "Vijay Thakur", 30000, "CS");
```

Emp_ID	Emp_Name	Salary	Dept
--------	----------	--------	------

INSERT INTO Employee2 VALUES (1, "Rishabh Dwivedi", 20000, "MCA");

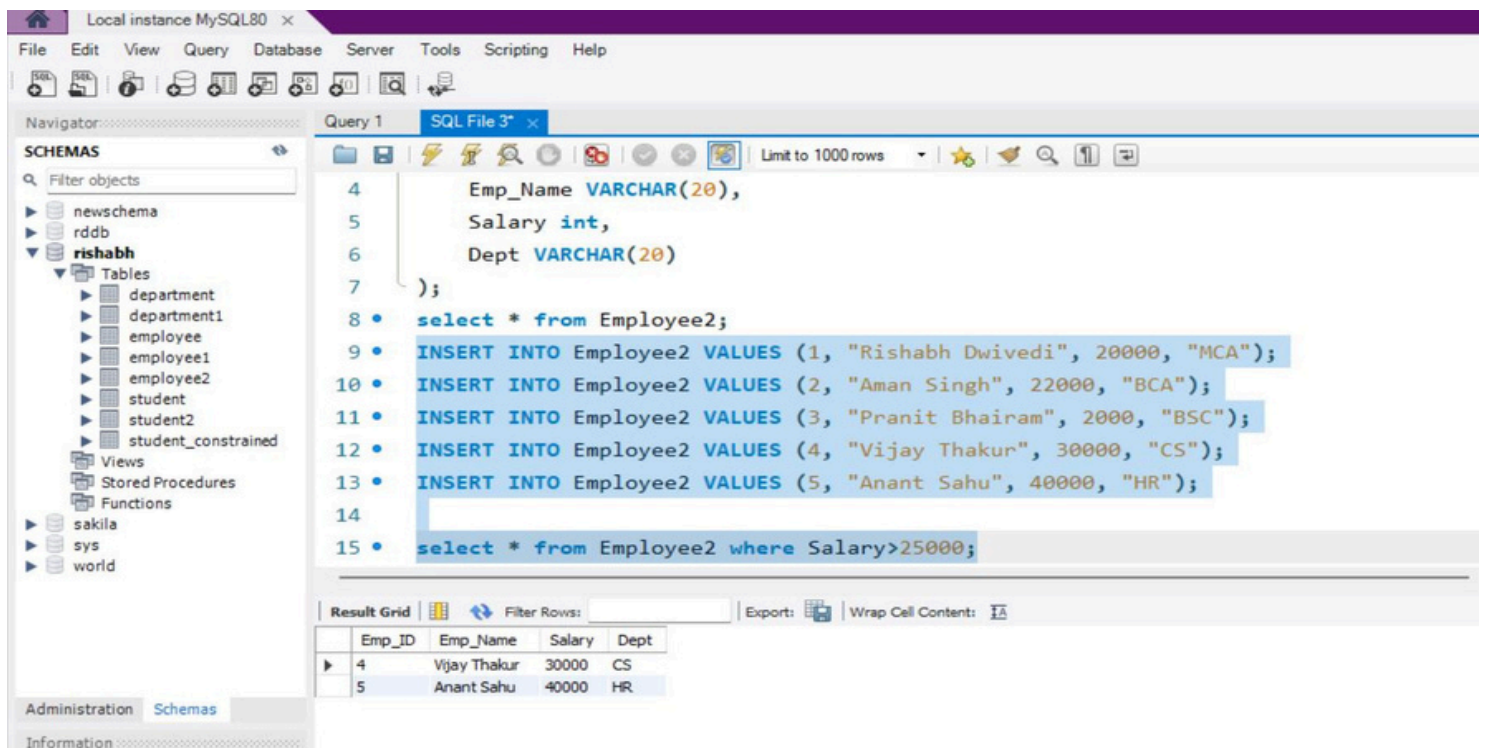
INSERT INTO Employee2 VALUES (2, "Aman Singh", 22000, "BCA");

INSERT INTO Employee2 VALUES (3, "Pranit Bhairam", 2000, "BSC");

INSERT INTO Employee2 VALUES (4, "Vijay Thakur", 30000, "CS");

INSERT INTO Employee2 VALUES (5, "Anant Sahu", 40000, "HR");

select * from Employee2 where Salary>25000;



The screenshot shows the MySQL Workbench interface with the same 'rishabh' database selected. The SQL script now includes five INSERT statements and a final SELECT query. The output of the SELECT query is displayed in the 'Result Grid' at the bottom, showing two rows where the salary is greater than 25000.

```
4     Emp_Name VARCHAR(20),
5     Salary int,
6     Dept VARCHAR(20)
7 );
8 select * from Employee2;
9 INSERT INTO Employee2 VALUES (1, "Rishabh Dwivedi", 20000, "MCA");
10 INSERT INTO Employee2 VALUES (2, "Aman Singh", 22000, "BCA");
11 INSERT INTO Employee2 VALUES (3, "Pranit Bhairam", 2000, "BSC");
12 INSERT INTO Employee2 VALUES (4, "Vijay Thakur", 30000, "CS");
13 INSERT INTO Employee2 VALUES (5, "Anant Sahu", 40000, "HR");
14
15 select * from Employee2 where Salary>25000;
```

Emp_ID	Emp_Name	Salary	Dept
4	Vijay Thakur	30000	CS
5	Anant Sahu	40000	HR

```

PL/SQL: CREATE TABLE Employee2 (
        Emp_ID NUMBER,
        Emp_Name VARCHAR2(20),
        Salary NUMBER,
        Dept VARCHAR2(20)
);
INSERT INTO Employee2 VALUES (1, 'Rishabh Dwivedi', 20000, 'MCA');
INSERT INTO Employee2 VALUES (2, 'Aman Singh', 22000, 'BCA');
INSERT INTO Employee2 VALUES (3, 'Pranit Bhairam', 2000, 'BSC');
INSERT INTO Employee2 VALUES (4, 'Vijay Thakur', 30000, 'CS');
INSERT INTO Employee2 VALUES (5, 'Anant Sahu', 40000, 'HR');
DECLARE
    CURSOR emp_cur IS
    SELECT Emp_Name, Salary FROM Employee2;
BEGIN
    FOR emp_rec IN emp_cur LOOP
        DBMS_OUTPUT.PUT_LINE(
            'EmployeeName: ' || emp_rec.Emp_Name || ' Salary: ' || emp_rec.Salary
        );
    END LOOP;
END;
/

```

queries.sql

+

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RUN

```

1  -- Create table
2  CREATE TABLE Employee2 (
3      Emp_ID  NUMBER,
4      Emp_Name VARCHAR2(20),
5      Salary  NUMBER,
6      Dept    VARCHAR2(20)
7  );
8  INSERT INTO Employee2 VALUES (1, 'Rishabh Dwivedi', 20000, 'MCA');
9  INSERT INTO Employee2 VALUES (2, 'Aman Singh', 22000, 'BCA');
10 INSERT INTO Employee2 VALUES (3, 'Pranit Bhairam', 2000, 'BSC');
11 INSERT INTO Employee2 VALUES (4, 'Vijay Thakur', 30000, 'CS');
12 INSERT INTO Employee2 VALUES (5, 'Anant Sahu', 40000, 'HR');
13 DECLARE
14 CURSOR emp_cur IS
15 SELECT Emp_Name, Salary FROM Employee2;
16 BEGIN
17 FOR emp_rec IN emp_cur LOOP
18 DBMS_OUTPUT.PUT_LINE(
19     'EmployeeName: ' || emp_rec.Emp_Name || ' Salary: ' || emp_rec.Salary
20 );
21 END LOOP;
22 END;
23 /
24

```

STDIN

Input for the program (Optional)

Output:

```

EmployeeName: Rishabh Dwivedi Salary: 20000
EmployeeName: Aman Singh Salary: 22000
EmployeeName: Pranit Bhairam Salary: 2000
EmployeeName: Vijay Thakur Salary: 30000
EmployeeName: Anant Sahu Salary: 40000

```


Practical 9

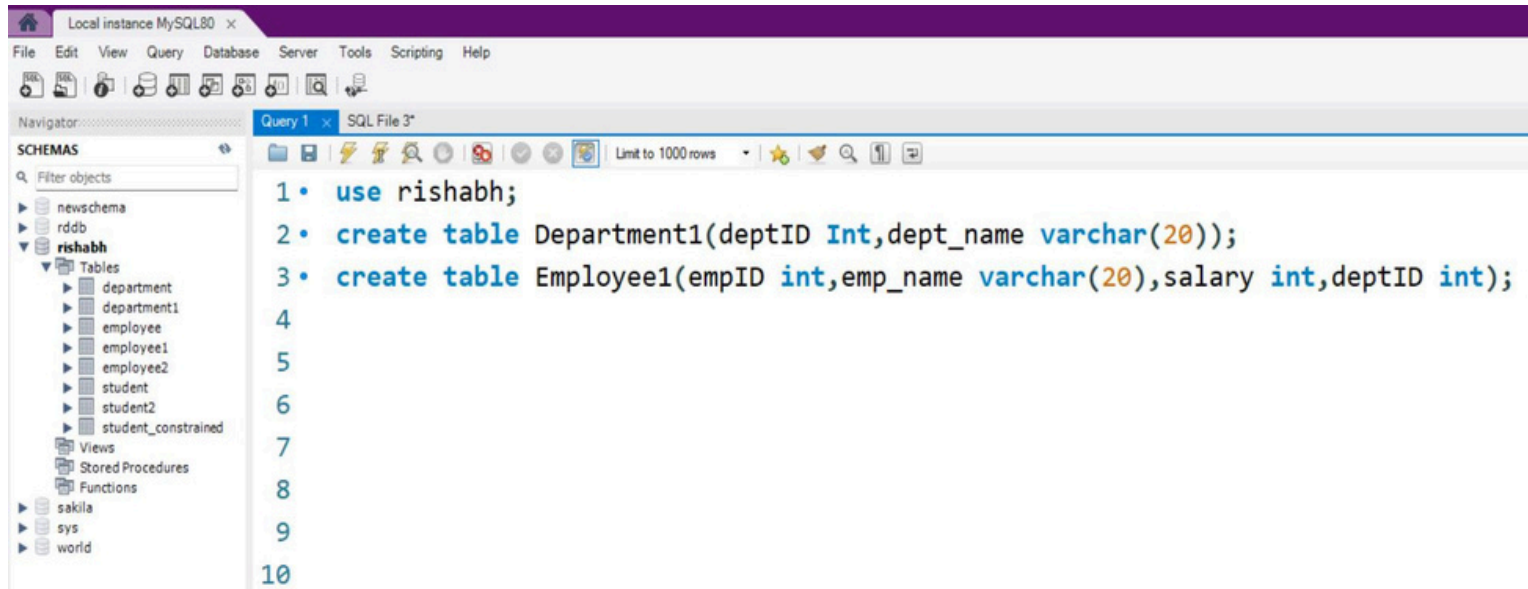
Aim: For a given set of relation schemes, create tables and perform the following Join Queries Inner Join, Outer Join Subqueries- With IN clause, With EXISTS clause.

Source-Code & Output:

use rishabh;

create table Department1(deptID int,dept_name varchar(20));

create table Employee1(empID int,emp_name varchar(20),salary int,deptID int);



insert into Department1 values(101,"MCA"),

(102,"BCA"),(103,"BSC"),

(104,"MSC"),(105,"HR"),(106,"MCA");

insert into Employee1 values(1,"Rishabh",10000,101),

(2,"Aman",20000,102),(3,"Pranit",5000,103),

(4,"Vijay",25000,104),(5,"Anant",15000,105),(6,"Sachin",35000,null);

	deptID	dept_name
▶	101	MCA
	102	BCA
	103	BSC
	104	MSC
	105	HR
	106	MCA

	empID	emp_name	salary	deptID
▶	1	Rishabh	10000	101
	2	Aman	20000	102
	3	Pranit	5000	103
	4	Vijay	25000	104
	5	Anant	15000	105
	6	Sachin	35000	NULL

select Employee1.emp_name,Department1.dept_namefrom Employee1

inner join Department1

on Employee1.deptID=Department1.deptID;

```

12 • select Employee1.emp_name,Department1.dept_name from Employee1
13 inner join Department1
14 on Employee1.deptID=Department1.deptID;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

emp_name	dept_name
Rishabh	MCA
Aman	BCA
Pranit	BSC
Vijay	MSC
Anant	HR

```

select Employee1.emp_name,Department1.dept_name from Employee1
left outer join Department1
on Employee1.deptID=Department1.deptID;

```

```

16 • select Employee1.emp_name,Department1.dept_name from Employee1
17 left outer join Department1
18 on Employee1.deptID=Department1.deptID;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

emp_name	dept_name
Rishabh	MCA
Aman	BCA
Pranit	BSC
Vijay	MSC
Anant	HR
Sachin	NULL

```

select Employee1.emp_name,Department1.dept_name from Employee1
right outer join Department1
on Employee1.deptID=Department1.deptID;

```

```

20 • select Employee1.emp_name,Department1.dept_name from Employee1
21 right outer join Department1
22 on Employee1.deptID=Department1.deptID;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

emp_name	dept_name
Rishabh	MCA
Aman	BCA
Pranit	BSC
Vijay	MSC
Anant	HR
NULL	MCA

```

SELECT emp_name, salary
FROM Employee1 WHERE deptID IN (
    SELECT deptID
    FROM Department1
    WHERE dept_name = 'MCA'
);

```

The screenshot shows a SQL IDE with a query editor and a result grid. The query is as follows:

```

24 • SELECT emp_name, salary
25 FROM Employee1 WHERE deptID IN (
26     SELECT deptID
27     FROM Department1
28     WHERE dept_name = 'MCA'
29 );

```

The result grid below the query shows the following data:

emp_name	salary
Rishabh	10000

```

SELECT emp_name
FROM Employee1 E WHERE EXISTS (
    SELECT *
    FROM Department1 D
    WHERE E.deptID = D.deptID
);

```

The screenshot shows a SQL IDE with a query editor and a result grid. The query is as follows:

```

31 • SELECT emp_name
32 FROM Employee1 E WHERE EXISTS (
33     SELECT *
34     FROM Department1 D
35     WHERE E.deptID = D.deptID
36 );

```

The result grid below the query shows the following data:

emp_name
Rishabh
Aman
Pranit
Vijay
Anant